E-Commerce Sales & Customer Insights - sql quries

Data Overview &Validation

- Loaded the dataset into SQL Server
- Previewed the data to understand structure and attributes
- Verified data consistency before performing analysis

```sql
select * from [dbo].[ecommerece]
```

Results | Messages

	Index	product_id	title	product_description	rating	ratings_count	initial_price	discount	final_price_1	currency	images
1	1	8376765	Lino Perros	Women Navy Blue Solid Backpack	3.79999995231628	15	3995	0.579999983310699	1677.90002441406	INR	http://assets.myntassets.com/assets/images/
2	2	9136281	Tommy Hilfiger	Unisex Navy Blue Striped Backpack	4.5	67	2899	0.349999994039536	1884.34997558594	INR	http://assets.myntassets.com/assets/images/
3	3	17633752	Lavie	Aries Women Pink Mini Backpack	4.40000009536743	226	2999	0.649999976158142	1049.65002441406	INR	http://assets.myntassets.com/assets/images/
4	4	1376949	F Gear	Unisex Navy & Grey Printed Burner Backpack	4.40000009536743	1052	1675	0.519999980926514	804	INR	http://assets.myntassets.com/assets/images/
5	5	13939916	MYTRIDENT	Men Blue Solid Bath Robe	4.69999980926514	12	2899	0.170000001788139	2406.169921875	INR	http://assets.myntassets.com/assets/images/
6	6	17198778	H&M	Beige & Tan Solid Stoneware Soap Dispenser	4.5	42	1399	0	1399	INR	http://assets.myntassets.com/assets/images/
7	7	19851824	AVI Living	Grey Solid 500 GSM Pure Cotton Bath Towel	4	5	1399	0.649999976158142	489.649993896484	INR	http://assets.myntassets.com/assets/images/
8	8	18602872	My Room	Yellow & Green Floral 140 TC King Bedsheet with 2 ...	4.69999980926514	10	2999	0.540000021457672	1379.5400390625	INR	http://assets.myntassets.com/assets/images/
9	409	15190770	AMERICAN EAGLE OUTFITTERS	Women Blue Slim Fit High-Rise Light Fade Jeans	4	6	3999	0.600000023841858	1599.59997558594	INR	http://assets.myntassets.com/assets/images/
10	410	22203852	Cantabil	Men Low Rise Clean Look Light Fade Stretchable C...	0	0	3999	0.660000026226044	1359.66003417969	INR	http://assets.myntassets.com/assets/images/
11	411	22005330	H&M	Women Slim Regular Ankle Jeans	3.79999995231628	5	2999	0	2999	INR	http://assets.myntassets.com/assets/images/
12	412	22203876	Cantabil	Men Cotton Heavy Fade Stretchable Jeans	0	0	3999	0.660000026226044	1359.66003417969	INR	http://assets.myntassets.com/assets/images/
13	413	15067844	Flying Machine	Men Blue Jackson Skinny Fit Low-Rise Heavy Fade ...	4.09999990463257	69	2399	0.5	1199.5	INR	http://assets.myntassets.com/assets/images/
14	414	19059688	Flying Machine	Men Blue Slim Fit Light Fade Stretchable Jeans	3.70000004768372	72	2999	0.550000011920929	1349.55004882813	INR	http://assets.myntassets.com/assets/images/
15	415	18960434	Lee	Men Blue Bruce Skinny Fit Light Fade Stretchable Je...	4.40000009536743	34	3999	0.550000011920929	1799.55004882813	INR	http://assets.myntassets.com/assets/images/

Top 5 Products by Sales

```sql
select top 5 title,
sum(initial_price) as total_sales
from [dbo].[ecommerece]
group by title
order by total_sales desc
```

- Identified top-performing products based on total sales
- Helps understand key revenue drivers

Results | Messages

	title	total_sales
1	ADIDAS	118784
2	U.S. Polo Assn.	98878
3	Levis	84279
4	KISAH	77895
5	Puma	67990

Product Rating & Risk Analysis

- Some high-selling products have low ratings
- Indicates potential quality or customer satisfaction issues

```sql
SELECT TOP 5 title, AVG(rating) avg_rating
FROM [dbo].[ecommerece]
GROUP BY title
ORDER BY avg_rating DESC;
```

Results | Messages

	title	avg_rating
1	ASICS	5
2	Bannos Swagger	5
3	Bruchi CLUB	5
4	DOOR74	5
5	ellementry	5

```sql
## top5 low rated products##
SELECT TOP 5 title, AVG(rating) avg_rating
FROM [dbo].[ecommerece]
GROUP BY title
ORDER BY avg_rating asc;
```

Results | Messages

	title	avg_rating
1	8seconds	0
2	Antony Morato	0
3	ASTRID	0
4	BELLA CASA	0
5	Braveo	0

```sql
##high sales low rating##
SELECT TOP 5 title,
SUM(initial_price) total_sales,
AVG(rating) avg_rating
FROM [dbo].[ecommerece]
GROUP BY title
HAVING AVG(rating) < 3
ORDER BY total_sales DESC;
```

Results | Messages

	title	total_sales	avg_rating
1	KISAH	77895	1.62000002861023
2	Saree mall	41991	1.75555552376641
3	Mitera	26894	1.88333332538605
4	Label Ritu Kumar	22500	2.82500001788139
5	MALHAAR	22273	0.528571435383388

Category-wise Sales Analysis

- A few categories contribute the majority of overall sales
- These categories represent high customer demand

```
#top 5 category wise sales#
select distinct top 5 category,title,
sum(initial_price)as total_sales
from[dbo].[ecommerece]
group by category,title
order by total_sales desc
```

Results | Messages

	category	title	total_sales
1	sports-shoes	ADIDAS	118784
2	sherwani	KISAH	77895
3	jeans	Levis	77282
4	jackets	U.S. Polo Assn.	72988
5	sports-shoes	Puma	56494

Price Range Analysis

- Medium-priced products contribute most to revenue
- Helps in pricing strategy decision

```sql
#price range analysis#
SELECT CASE
 WHEN initial_price < 500 THEN 'Low'
 WHEN initial_price <= 2000 THEN 'Medium'
 ELSE 'High' END AS price_range,
 SUM(initial_price) total_sales
FROM [dbo].[ecommerece]
GROUP BY CASE
 WHEN initial_price < 500 THEN 'Low'
 WHEN initial_price <= 2000 THEN 'Medium'
 ELSE 'High' END;
```

Results | Messages

	price_range	total_sales
1	High	2026986
2	Low	16230
3	Medium	680025

Key Business Insights

- Top products and categories contribute the majority of total sales
- Some high-selling products have low customer ratings
- Customer satisfaction varies across products
- Medium-priced products generate the highest revenue
- Low-rated products indicate potential quality issues

Business Recommendations

- Improve quality of high-sales but low-rated products
- Focus marketing on top-performing categories
- Promote high-rated but low-selling products
- Optimize pricing strategy for better revenue
- Monitor customer feedback regularly